

SET-C | Quiz-2 | CSE425.11 | NSU | Date: 25.7.26

1. Which compiler phase scans the source program and groups characters into meaningful tokens, discarding whitespace and comments in the process?

- A. Parsing
- B. Lexical Analysis
- C. Semantic Analysis
- D. Optimization

2. What is produced when the Parsing phase processes a stream of tokens according to the language's grammar?

- A. A symbol table
- B. Optimized machine code
- C. A parse tree (syntax tree)
- D. A list of lexical errors only

3. Checking that operators are applied to compatible operand types, such as flagging "string + integer", is performed during:

- A. Lexical Analysis
- B. Parsing
- C. Semantic Analysis
- D. Code Generation

4. Which phase transforms the intermediate representation to reduce running time or resource usage while preserving the program's original meaning?

- A. Parsing
- B. Optimization
- C. Semantic Analysis
- D. Lexical Analysis

5. Which phase is responsible for producing the final target-machine instructions from the (optimized) intermediate representation?

- A. Code Generation
- B. Parsing
- C. Semantic Analysis
- D. Lexical Analysis

6. Which of the following correctly lists the five phases of a compiler in their proper sequence?

- A. Lexical Analysis → Semantic Analysis → Parsing → Code Generation → Optimization
- B. Parsing → Lexical Analysis → Optimization → Semantic Analysis → Code Generation
- C. Lexical Analysis → Parsing → Semantic Analysis → Optimization → Code Generation
- D. Code Generation → Optimization → Semantic Analysis → Parsing → Lexical Analysis

7. What does the shorthand character class \D match in a regex?

- A. Any digit character
- B. Any non-digit character
- C. The digit range 0-9 only
- D. Any decimal point

8. Which pattern matches any single alphanumeric character (a letter, digit, or underscore)?

- A. \W
- B. \s
- C. \w
- D. \S

9. Which pattern matches any character that is NOT alphanumeric?

- A. \w
- B. \W
- C. \d
- D. \D

10. The pattern [0-9] matches:

- A. Any letter from a to z
- B. Any single digit from 0 to 9
- C. The literal string "0-9"
- D. Any punctuation mark

11. What does the pattern [xyz] match?

- A. The exact sequence "xyz"
- B. Any one of the characters x, y, or z
- C. Any character except x, y, or z
- D. Zero or more occurrences of x, y, z

12. What does the pattern [^0-9] match?

- A. Any digit from 0 to 9
- B. Any character that is not a digit
- C. The literal string "^0-9"
- D. Any digit immediately followed by a caret

13. In regular expressions, \S matches:

- A. Any whitespace character
- B. Any non-whitespace character
- C. A literal letter 'S'
- D. Any digit character

14. Since ? is a regex metacharacter, which pattern correctly matches a literal question mark character?

- A. ?
- B. \?
- C. ^?
- D. ?*

15. Which pattern matches only the exact standalone word "dog" and not "dogs" or "hotdog"?

- A. dog
- B. ^dog\$
- C. .dog.
- D. [dog]

16. What does the pattern [a-zA-Z]{2,4} mean?

- A. Exactly 2 letters followed by 4 more letters
- B. Between 2 and 4 repetitions of a letter (upper or lower case)
- C. At least 4 letters
- D. Exactly 24 letters in a row

17. Which metacharacter matches one or more repetitions of the preceding element?

- A. *
- B. ?
- C. +
- D. {0,1}

18. Which metacharacter makes the preceding element match exactly zero or one time?

- A. +
- B. *
- C. ?
- D. {1}

19. Given the pattern (cat|dog), which strings would it match?

- A. Only "catdog"
- B. Either "cat" or "dog"
- C. "cat" followed by "dog" repeated
- D. Neither "cat" nor "dog"

20. In the pattern ((ab)c), what does the inner group (ab) represent?

- A. An unrelated, independent match
- B. A nested capture group that captures "ab" within the larger match "abc"
- C. An optional part of the pattern
- D. A character class excluding a and b